

Harthik Royal Mallichetty

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PROFESSIONAL SUMMARY

Quantitative Research Analyst and May 2026 MSBA graduate with hands-on experience building financial models, computing institutional-grade risk metrics, and communicating rigorous quantitative findings to stakeholders. Built Quant Risk Desk, a production pipeline computing VaR/CVaR, Sharpe/Sortino, rolling Beta, and Macro Regime classification across 11 S&P 500 sector ETFs. Previously built a production predictive model at Prolifics achieving 89%+ accuracy, still running in 2025. Proficient in Python, SQL, R, dbt, PostgreSQL, Tableau, and statistical modeling. Open to relocation.

EDUCATION

The University of Texas at Dallas, Richardson, Texas

Master of Science in Business Analytics and Artificial Intelligence – Scholar With Distinction

May 2026

GPA 3.81

GITAM University, Hyderabad, India

April 2024

Bachelor of Technology in Computer Science and Business Systems

GPA 3.73

PROFESSIONAL EXPERIENCE

Prolifics – Data & Analytics Intern | Hyderabad, India

Feb 2023 – Jan 2024

- Built regression model achieving 89%+ accuracy on sprint outcome prediction; designed SQL pipeline ingesting 100K+ PostgreSQL records, deployed Flask REST API integrated with company portal used by 200 developers daily.
- Analyzed complex datasets using Python and SQL to discover performance patterns; automated ETL workflows processing 50K+ rows/day reducing reporting cycle time by 40% and improving accuracy by 25%.

HeiDigi – Digital Marketing Analyst Intern | Hyderabad, India

Mar 2022 – Aug 2022

- Applied statistical analysis and hypothesis testing to design A/B tests evaluating campaign performance; translated quantitative findings into actionable recommendations improving ROI by 20% and CTR by 18%.

PROJECTS

Quant Risk Desk | Python, SQL, dbt, Apache Airflow, PostgreSQL, Docker, Tableau

April 2026

- Developed quantitative financial models computing rolling VaR/CVaR at 95% and 99% confidence intervals, annualized Sharpe/Sortino ratios, and 252-day rolling Beta across 11 S&P 500 sector ETFs; designed Macro Regime Engine classifying market conditions using Fed Funds Rate, CPI, VIX, yield curve inversion, and high-yield spread thresholds.
- Identified diversification collapse from 0.47 to 0.88 correlation during March 2020 COVID crash across 182K+ pair-day observations; implemented 3-layer dbt transformation with 18 tests enforcing zero violations across 21K+ records.

Signal – Financial Intelligence | Python, pgvector, dbt, Airflow, Voyage AI, GPT-4o-mini, GCP

May 2026

- Designed dbt 3-layer transformation computing 65 financial metrics per company across 70 S&P 500 companies including revenue growth, operating margins, YoY trends, and TTM figures; marts layer serves as sole API data source ensuring no raw table queries hit production endpoints.
- Built financial Q&A system where investors ask questions in plain English and receive cited answers from SEC filings in seconds; 10-step RAG pipeline with GPT-4o-mini query classification, Voyage AI finance embeddings, and pgvector HNSW retrieval running on GCP.

Global Supply Chain Intelligence | Python, SQL, XGBoost, LightGBM, SHAP, NetworkX, Plotly

Mar 2026

- Executed Monte Carlo simulation with 10K runs per SKU across 3 geopolitical stress scenarios quantifying financial exposure: baseline \$13.8M, moderate \$18.1M, severe \$19.1M; applied SHAP TreeExplainer attributing 59% of model weight to disruption signals across XGBoost and LightGBM ensemble (Recall 70.9%).
- Built CUSUM and Mahalanobis anomaly detection achieving 100% disruption recall with 8-week lead time across 78K weekly records and 500 SKUs; delivered 6-tab interactive Plotly dashboard communicating quantitative findings to non-technical stakeholders.

SKILLS

Quantitative and Statistical Methods: Financial Modeling, VaR/CVaR (Historical Simulation), Sharpe/Sortino Ratios, Rolling Beta, Macro Regime Classification, Monte Carlo Simulation, CUSUM Detection, Hypothesis Testing, A/B Testing, Regression Analysis, Time-Series Forecasting (ARIMA, ETS, Prophet), Champion-Challenger Modeling

Programming and Tools: Python (Pandas, NumPy, scikit-learn, XGBoost, SHAP, SciPy, matplotlib, Plotly), R (statistical analysis, econometrics, time-series), SQL (CTEs, Window Functions, Self-Joins, Rolling Aggregations), Git/GitHub

Data Engineering: dbt (Staging/Intermediate/Marts, custom tests), Apache Airflow, PostgreSQL, DuckDB, Docker, ETL Pipeline Automation, Data Validation and Quality Testing, Dimensional Modeling

Analytics and Visualization: Tableau, Power BI (DAX, Data Modeling), Excel (Advanced Financial Modeling, Forecasting, Solver), Streamlit, Plotly, Data Storytelling, Executive Presentations

Domain Knowledge: Financial Risk Modeling, Market Regime Analysis, Credit Risk, Supply Chain Risk Analytics, Quantitative Research, SEC Filing Analysis

ACTIVITIES

Cloud Analytics Lead – GDSC, GITAM Hyderabad

Apr 2022 – Apr 2024

- Built GCP and BigQuery pipelines analyzing behavioral data to quantify investment impact on college admission rates.